

VENDOR PROCUREMENT · NETWORK INFRASTRUCTURE

# Enterprise Campus Network Monitoring System

This document defines our requirements for the procurement of an enterprise-grade network monitoring platform. Responding vendors are expected to demonstrate full alignment with all Must Have requirements specified herein.

|                  |                                                         |
|------------------|---------------------------------------------------------|
| Document Title   | Enterprise Campus Network Monitoring System — BRD       |
| Reference No.    | NMS-BRD-2026                                            |
| Version          | 2.0                                                     |
| Date             | June 2026                                               |
| Status           | For Vendor Distribution                                 |
| Issued By        | Network Procurement Office — IT Infrastructure Division |
| Deployment Scope | On-Premises · Up to 1,000 Nodes · Multi-Site Campus     |
| Compliance       | NCA ECC 2018 · ISO/IEC 27001                            |
| Classification   | Confidential — Vendor Use Only                          |

CONFIDENTIAL NOTICE

This document is intended solely for authorized vendors participating in our network monitoring platform procurement process. Recipients must not disclose, reproduce, or distribute this document to any third party without prior written consent.

DOCUMENT

Table of Contents

BACKGROUND

|   |                                      |   |
|---|--------------------------------------|---|
| 1 | Executive Summary                    | 3 |
| 2 | Organization & Current Environment   | 3 |
| 3 | Business Problem Statement           | 3 |
| 4 | Project Objectives & Success Metrics | 4 |
| 5 | Scope                                | 4 |
| 6 | Stakeholders                         | 4 |

REQUIREMENTS

|    |                                      |     |
|----|--------------------------------------|-----|
| 7  | Functional Requirements              | 5–7 |
| 8  | Non-Functional Requirements          | 8   |
| 9  | Technical Environment                | 9   |
| 10 | Compliance & Regulatory Requirements | 9   |

PROCUREMENT

|    |                                       |    |
|----|---------------------------------------|----|
| 11 | Vendor Evaluation Criteria            | 10 |
| 12 | Commercial & Licensing Requirements   | 10 |
| 13 | Implementation & Support Requirements | 11 |
| 14 | Acceptance Criteria                   | 11 |
| 15 | Risks & Constraints                   | 12 |
| 16 | Vendor Response Instructions          | 12 |

APPENDICES

|   |              |    |
|---|--------------|----|
| A | Priority Key | 13 |
| B | Glossary     | 13 |

DOCUMENT HISTORY

Revision History

| VERSION | DATE      | AUTHOR                 | DESCRIPTION                                                                                                                                         |
|---------|-----------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | May 2026  | Business Analysis Team | Initial internal draft                                                                                                                              |
| 2.0     | June 2026 | IT Procurement Team    | Revised for vendor distribution — reframed as procurement BRD; added evaluation criteria, commercial requirements, and vendor response instructions |

PRIORITY KEY SUMMARY

Requirements in this document are classified as **Must Have (M)** — disqualifying if not met; **Should Have (S)** — weighted in scoring; **Could Have (C)** — low-weight differentiator. See Appendix A for full definitions.

1

SECTION 1

Executive Summary

We operate a large-scale enterprise campus network serving clinical, administrative, and research functions. We are procuring an enterprise-grade network performance monitoring platform to strengthen our network visibility and operational capabilities.

This BRD defines the full scope of functional requirements, non-functional requirements, integration needs, compliance obligations, and commercial expectations. Vendors are invited to respond with a formal proposal demonstrating how their solution meets each stated requirement. The selected platform must provide complete, real-time visibility into network infrastructure health, performance, and availability from a centralized, web-based console deployed on-premises within our on-premises data center.

2

SECTION 2

Organization & Current Environment

2.1

About the Organization

We are a healthcare institution operating an enterprise-grade network spanning campus buildings, data centers, and clinical zones. Our network infrastructure is primarily Cisco. All network devices have SNMP enabled with standardized credentials across the estate.

2.2

Current Monitoring Environment

| ATTRIBUTE           | CURRENT STATE                                                                    |
|---------------------|----------------------------------------------------------------------------------|
| Approx. Scale       | ~1,000 monitored nodes (current); growth target to be confirmed within 24 months |
| Sites               | 1 primary campus site; 1 remote WAN-connected site                               |
| Deployment Model    | On-premises, Windows Server                                                      |
| Active Integrations | Email notifications, SNMP traps (no Active Directory/LDAP currently deployed)    |
| Primary Users       | NOC team, Network Engineering, IT Management                                     |

2.3

Drivers for Change

Operational Visibility

A unified monitoring platform is essential to maintaining proactive visibility across our campus network and reducing MTTR.

TCO Reduction

Lower total cost of ownership over a 3-year horizon is a key selection criterion.

Vendor Independence

Eliminate single-vendor lock-in for network observability infrastructure.

Long-Term Roadmap

Require a vendor with demonstrated product stability and an active 3-year development roadmap.

3

SECTION 3

Business Problem Statement

Without a unified, real-time monitoring platform, our IT Operations team faces the following operational risks:

| PROBLEM AREA               | IMPACT                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------|
| Reactive incident response | Outages detected by end-user complaints rather than proactive alerting — extends MTTR         |
| Monitoring blind spots     | Multi-vendor infrastructure produces fragmented visibility with no single pane of glass       |
| No root cause correlation  | Performance degradation cannot be traced across L2/L3 without automated dependency mapping    |
| Absent capacity planning   | Bandwidth saturation and hardware refresh decisions are reactive without trending data        |
| Compliance reporting gaps  | Uptime SLAs, audit trails, and availability metrics must be generated manually                |
| No traffic visibility      | Cannot identify top bandwidth consumers, rogue applications, or abnormal traffic in real time |



4

SECTION 4

Project Objectives & Success Metrics

COVERAGE

≥98%

Managed devices discovered within 48 hrs of deployment

ALERT LATENCY

≤60s

Threshold breach to notification delivery

MTTR REDUCTION

≥35%

vs. baseline within 90 days of go-live

CONSOLE UPTIME

99.9%

Platform availability SLA for monitoring console

NOC ADOPTION

100%

NOC operators trained and live within 30 days

CUTOVER GAP

0

Zero monitoring gap during platform go-live and cutover

5

SECTION 5

Scope

5.1

In Scope

- Device discovery, inventory & classification (SNMP, ICMP, CDP/LLDP)
- Real-time SNMP polling — CPU, memory, interface metrics
- NetFlow / sFlow / IPFIX traffic collection & analysis
- Layer 2 & Layer 3 topology discovery & mapping
- Threshold-based & anomaly alerting with escalation
- Drag-and-drop dashboards & pre-built NOC views
- Scheduled reporting (PDF, Excel, HTML)
- RBAC with local user account management
- REST API for ITSM integration
- High-availability (HA) deployment architecture
- Cisco device support (primary infrastructure); vendor must confirm multi-vendor extensibility for future use

5.2

Out of Scope

- Application Performance Monitoring (APM)
- Log aggregation / SIEM / SOAR
- Configuration backup & change management (NCM)
- Endpoint or agent-based security monitoring
- SD-WAN / SASE-specific monitoring
- Mobile native applications

5.3

Key Constraints

|                |                                                   |
|----------------|---------------------------------------------------|
| Deployment     | On-premises only — no cloud SaaS                  |
| Data Residency | All data stays within our on-premises environment |
| Compliance     | NCA ECC 2018 mandatory                            |
| Transition     | No monitoring gap during cutover                  |

6

SECTION 6

Stakeholders

| Stakeholder              | Role                    | Responsibility                                                             | Engagement |
|--------------------------|-------------------------|----------------------------------------------------------------------------|------------|
| IT Operations Director   | Primary Sponsor         | Approve requirements and budget; final sign-off                            | High       |
| Network Engineering Team | Subject Matter Expert   | Validate monitoring requirements and device coverage                       | High       |
| NOC Manager              | End User Representative | Define alerting workflows, dashboard needs, NOC workflows                  | High       |
| IT Security Team         | Reviewer                | Validate access control, data security, compliance requirements            | Medium     |
| Procurement Team         | Evaluation Coordinator  | Coordinate vendor evaluation, requirement alignment, and selection process | High       |
| Executive Leadership     | Executive Sponsor       | Final investment decision and budget governance                            | Low        |

7

SECTION 7

Functional Requirements

All requirements are tagged with priority: Must Have Should Have Could Have . Must Have items are disqualifying if not natively supported. See Appendix A for full definitions.

7.1

Device Discovery & Inventory

| ID    | REQUIREMENT                                                                                                        | PRIORITY    | NOTES                          |
|-------|--------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------|
| FR-01 | Automated ICMP ping sweep across user-defined IP ranges or CIDR subnets to discover all live hosts                 | Must Have   | CIDR notation support required |
| FR-02 | Poll and manage devices via SNMP v1, v2c, and v3 with configurable community strings and v3 auth/priv credentials  | Must Have   |                                |
| FR-03 | Use CDP and LLDP to discover adjacent devices and build neighbor tables automatically                              | Must Have   |                                |
| FR-04 | Import and parse custom MIB files to extend monitoring to proprietary or niche device types                        | Should Have | MIB browser preferred          |
| FR-05 | Auto-classify discovered devices by type: router, switch, firewall, server, wireless AP, printer, UPS              | Must Have   |                                |
| FR-06 | Schedule recurring discovery scans (hourly, daily, weekly) with delta reporting of new/removed devices             | Must Have   |                                |
| FR-07 | Live device inventory view with filterable columns: device name, IP, type, vendor, OS version, status, last polled | Must Have   |                                |
| FR-08 | Bulk device import via CSV for large-scale on-boarding                                                             | Should Have |                                |

7.2

Real-Time Performance Monitoring

| ID    | REQUIREMENT                                                                                                     | PRIORITY    | NOTES                              |
|-------|-----------------------------------------------------------------------------------------------------------------|-------------|------------------------------------|
| FR-10 | Poll device CPU and memory utilization via SNMP at configurable intervals (default 60s; minimum 30s)            | Must Have   |                                    |
| FR-11 | Monitor interface-level metrics per port: bandwidth utilization (in/out), error rate, discard rate, packet loss | Must Have   | Per-interface granularity required |
| FR-12 | Real-time up/down status with color-coded indicators: green (up), red (down), yellow (warning), grey (unknown)  | Must Have   |                                    |
| FR-13 | Calculate and display device response time (latency) via ICMP round-trip time per device                        | Must Have   |                                    |
| FR-14 | Retain raw performance data minimum 90 days; aggregated hourly/daily data minimum 2 years                       | Must Have   | Configurable retention policy      |
| FR-15 | WMI polling for Windows servers: disk I/O, service status, process CPU                                          | Should Have |                                    |
| FR-16 | SNMP trap reception and correlation with polled device state                                                    | Must Have   |                                    |
| FR-17 | Interface utilization trend sparklines or heat maps within device detail views                                  | Should Have |                                    |

7.3

Network Traffic Analysis (Flow)

| ID    | REQUIREMENT                                                                                                  | PRIORITY    | NOTES                        |
|-------|--------------------------------------------------------------------------------------------------------------|-------------|------------------------------|
| FR-20 | Receive and parse NetFlow v5/v9, sFlow v5, and IPFIX records (UDP listener, configurable port, default 2055) | Must Have   |                              |
| FR-21 | Classify traffic by application using NBAR / DPI signatures; display top-N applications by bandwidth         | Must Have   |                              |
| FR-22 | Display top talkers by source IP, destination IP, protocol, and port with conversation-level drill-down      | Must Have   |                              |
| FR-23 | Historical flow analysis with selectable time ranges: 1h, 6h, 24h, 7d, 30d, and custom range                 | Must Have   |                              |
| FR-24 | Alert on bandwidth utilization thresholds per interface or aggregate circuit capacity                        | Must Have   | Integrates with alert engine |
| FR-25 | Per-VLAN and per-subnet traffic breakdown                                                                    | Should Have |                              |
| FR-26 | Identify and flag anomalous traffic patterns (e.g., port scans, sudden bandwidth spikes)                     | Should Have |                              |

7.4

Topology Discovery & Mapping

| ID    | REQUIREMENT                                                                                       | PRIORITY    | NOTES |
|-------|---------------------------------------------------------------------------------------------------|-------------|-------|
| FR-30 | Auto-generate Layer 2 topology maps using CDP/LLDP neighbor discovery data without manual input   | Must Have   |       |
| FR-31 | Generate Layer 3 routing topology maps using OSPF/BGP tables retrieved via SNMP                   | Should Have |       |
| FR-32 | Overlay live device health status on topology maps using color-coded status indicators            | Must Have   |       |
| FR-33 | Custom network maps with drag-and-drop device placement and optional background floor plan images | Must Have   |       |
| FR-34 | Export topology maps as PNG, PDF, or SVG                                                          | Should Have |       |
| FR-35 | Link utilization levels shown on map edges with color graduation (green/amber/red)                | Must Have   |       |
| FR-36 | Topology map drill-down — clicking a device opens its detail/performance view                     | Must Have   |       |

7.5

Alerting & Notification Engine

| ID    | REQUIREMENT                                                                                                         | PRIORITY    | NOTES                      |
|-------|---------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|
| FR-40 | Threshold-based alerts on any monitored metric: CPU %, memory %, interface utilization, packet loss, device up/down | Must Have   |                            |
| FR-41 | Multi-condition alert rules using AND/OR boolean logic across multiple metrics and device groups                    | Must Have   |                            |
| FR-42 | Notifications via: email (SMTP), SMS (HTTP gateway), webhook (outbound JSON POST), SNMP trap forwarding             | Must Have   | All four channels required |
| FR-43 | Alert escalation policies — auto-escalate to next tier if unacknowledged within configurable time window            | Should Have |                            |
| FR-44 | Maintenance windows (one-time and recurring) that automatically suppress alerts for planned downtime                | Must Have   |                            |
| FR-45 | Dependent alert suppression — suppress child device alerts when upstream parent is confirmed down                   | Must Have   | Root cause suppression     |
| FR-46 | Alert acknowledgement, annotation, and assignment to operators from within the console                              | Must Have   |                            |
| FR-47 | Alert history log with search, filter, and export capability                                                        | Must Have   |                            |
| FR-48 | Alert deduplication to prevent notification flooding for flapping devices                                           | Must Have   |                            |

7.6

Dashboards & Reporting

| ID    | Requirement                                                                                                                   | Priority    | Notes                |
|-------|-------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|
| FR-50 | Drag-and-drop dashboard builder with widget types: gauge, line chart, bar chart, table, topology tile, status summary         | Must Have   |                      |
| FR-51 | Out-of-the-box dashboards: Executive Summary, NOC Real-Time Overview, Interface Detail, Top-10 Bandwidth Users, Device Health | Must Have   |                      |
| FR-52 | Scheduled reports delivered via email in PDF, Excel, and HTML formats                                                         | Must Have   |                      |
| FR-53 | SLA availability reporting per device or group: uptime %, MTTR, MTBF, downtime events over configurable date ranges           | Must Have   |                      |
| FR-54 | Capacity planning reports that trend current utilization and project when thresholds will be exceeded                         | Should Have | Linear trend minimum |
| FR-55 | Report templates saveable and shareable across user accounts                                                                  | Should Have |                      |
| FR-56 | Dashboard sharing via read-only public URL or embedded iFrame (NOC wall display use case)                                     | Should Have |                      |

7.7

Access Control & User Management

| ID    | Requirement                                                                                                                 | Priority   | Notes                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------|
| FR-60 | RBAC with built-in roles: Administrator, Operator, Read-Only; plus custom role creation with granular permission assignment | Must Have  |                                                 |
| FR-61 | Integration with Microsoft Active Directory and LDAP for user authentication and group-to-role mapping                      | Could Have | Not currently deployed; future requirement only |
| FR-62 | SAML 2.0 for single sign-on (SSO) integration                                                                               | Could Have | Not currently required                          |
| FR-63 | Organizational unit-based device visibility scoping — users only see devices within their assigned scope                    | Must Have  |                                                 |
| FR-64 | Tamper-evident audit log of all user actions: login/logout, config changes, alert acknowledgements, report generation       | Must Have  |                                                 |
| FR-65 | Account lockout policy after configurable number of failed login attempts                                                   | Must Have  |                                                 |

7.8

API & External Integration

| ID    | Requirement                                                                                                  | Priority    | Notes |
|-------|--------------------------------------------------------------------------------------------------------------|-------------|-------|
| FR-70 | Documented REST API (JSON) for programmatic read/write access to devices, alerts, metrics, and configuration | Must Have   |       |
| FR-71 | Outbound webhook triggers on alert state changes (open, acknowledged, resolved)                              | Must Have   |       |
| FR-72 | Native or documented integration with ServiceNow for automatic incident creation on critical alerts          | Should Have |       |
| FR-73 | Receive syslog messages from network devices and correlate with polled device state                          | Should Have |       |
| FR-74 | SNMP trap reception from managed devices                                                                     | Must Have   |       |



8

SECTION 8

Non-Functional Requirements

8.1

Performance

| ID     | REQUIREMENT                                                          | TARGET / METRIC   |
|--------|----------------------------------------------------------------------|-------------------|
| NFR-01 | Dashboard load time for standard views with up to 500 devices        | < 3 seconds       |
| NFR-02 | Full polling cycle duration for 10,000 monitored interfaces          | < 60 seconds      |
| NFR-03 | NetFlow / sFlow / IPFIX record ingestion and processing throughput   | ≥ 100,000 flows/s |
| NFR-04 | Alert trigger latency from threshold breach to notification delivery | < 60 seconds      |
| NFR-05 | Search / filter response across full device inventory                | < 2 seconds       |

8.2

Scalability

| ID     | REQUIREMENT                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------|
| NFR-10 | Support at least <b>5,000 monitored nodes</b> and <b>100,000 monitored interfaces</b> in a single deployment          |
| NFR-11 | Distributed polling architecture must support horizontal scaling by adding polling engines with no platform reinstall |
| NFR-12 | Database architecture must maintain query performance as historical data grows beyond 1 TB                            |
| NFR-13 | Support at least <b>50 concurrent authenticated web console users</b> without performance degradation                 |

8.3

Availability & Reliability

| ID     | REQUIREMENT                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------|
| NFR-20 | Platform shall maintain <b>99.9% uptime</b> for the monitoring console (< 8.76 hours unplanned downtime/year)        |
| NFR-21 | Hot-standby HA configuration must achieve failover within <b>60 seconds</b> with no data loss                        |
| NFR-22 | All configuration and historical data backed up daily with minimum 30-day backup retention                           |
| NFR-23 | Distributed polling engines must continue collecting data independently if central server is temporarily unreachable |

8.4

Security

| ID     | REQUIREMENT                                                                                                        |
|--------|--------------------------------------------------------------------------------------------------------------------|
| NFR-30 | All web console traffic served over HTTPS (TLS 1.2 minimum; TLS 1.3 preferred)                                     |
| NFR-31 | SNMP v3 credentials stored encrypted at rest (AES-256 minimum)                                                     |
| NFR-32 | User passwords stored using strong one-way hashing algorithm (bcrypt or Argon2)                                    |
| NFR-33 | Session timeout after 30 minutes of inactivity (administrator-configurable)                                        |
| NFR-34 | Third-party OWASP Top 10 vulnerability assessment completed within the past 24 months; report available on request |
| NFR-35 | No monitoring data, device credentials, or telemetry transmitted to any external cloud service                     |

8.5

Usability

| ID     | REQUIREMENT                                                                                                        |
|--------|--------------------------------------------------------------------------------------------------------------------|
| NFR-40 | Web console fully functional on Chrome, Firefox, Edge (Chromium), and Safari (latest 2 major versions)             |
| NFR-41 | UI responsive and functional at 1280×800 minimum resolution                                                        |
| NFR-42 | New administrator able to complete initial device discovery and receive first alert within 4 hours of installation |
| NFR-43 | In-product contextual help or searchable knowledge base accessible from within the console                         |



9

SECTION 9

Technical Environment

9.1

Current Network Infrastructure

| CATEGORY         | DETAILS                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------|
| Network Vendors  | Cisco (all network devices)                                                                   |
| Protocols in Use | SNMP v2c/v3, OSPF, BGP, CDP, LLDP                                                             |
| Flow Export      | NetFlow v9 (Cisco)                                                                            |
| SNMP Readiness   | 100% of devices SNMP-enabled; standardized credentials; all versions (v1/v2c/v3) supported    |
| Sites            | 1 primary campus site; 1 remote WAN-connected site requiring proxy or buffered polling config |
| Authentication   | Local accounts (no Active Directory or LDAP currently deployed)                               |
| ITSM Platform    | ServiceNow                                                                                    |
| Server OS        | Windows Server 2019/2022, RHEL 8/9, Ubuntu 22.04 LTS                                          |

9.2

Deployment Requirements

DEPLOYMENT MODEL

On-premises only. Cloud-connected SaaS models are **not acceptable**. A VM is already provisioned. Vendor must provide resource specifications (vCPU, RAM, storage) suited for ~1,000 nodes initial scale with a clear upgrade path.

| REQUIREMENT       | DETAILS                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Supported OS      | Windows Server 2019/2022 and/or Linux (RHEL 8/9 or Ubuntu 22.04 LTS)                                                                   |
| Virtualization    | VMware vSphere — VM pre-provisioned; vendor to provide resource specifications (vCPU, RAM, storage) based on 1,000-node starting scale |
| High Availability | Active-passive HA required — vendor must describe architecture in proposal                                                             |
| Database          | Vendor must describe embedded or external DB engine and provide storage sizing guidance                                                |

9.3

Integration Points

| SYSTEM                  | INTEGRATION TYPE      | REQUIREMENT                                                                  |
|-------------------------|-----------------------|------------------------------------------------------------------------------|
| Active Directory / LDAP | LDAP / LDAPS          | Not currently deployed — future integration capability required (Could Have) |
| ServiceNow              | REST webhook outbound | Auto-create incidents from critical alerts                                   |
| Email Server            | SMTP with TLS         | Alert notifications and scheduled report delivery                            |
| SMS Gateway             | HTTP API              | Critical alert SMS via third-party gateway                                   |
| Syslog                  | UDP/TCP               | Receive device syslog events                                                 |
| External APIs           | REST (JSON)           | Programmatic read access for third-party tooling                             |

10

SECTION 10

Compliance & Regulatory Requirements

All aspects of the solution — data storage, access control, encryption, and audit logging — must comply with the following. Vendors must explicitly state compliance posture against each item in their proposal.

| STANDARD / REGULATION    | RELEVANCE & EXPECTATION                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCA ECC 2018 (KSA)       | Saudi National Cybersecurity Authority Essential Controls — governs monitoring tool data handling, access management, and encryption. <b>Mandatory compliance.</b> |
| HIPAA (where applicable) | Our organization handles patient data. Network monitoring systems must not inadvertently capture or expose PHI. Vendor must confirm scope.                         |
| ISO/IEC 27001            | Vendor must confirm that their development and delivery processes are ISO 27001 certified or in active pursuit. Certificate or status letter required.             |
| GDPR (informational)     | Applicable to any data processed on behalf of non-KSA entities. Vendor must clarify data processing scope and confirm no data leaves the on-prem environment.      |

11

SECTION 11

Vendor Evaluation Criteria

| # | CRITERION                                                                                                                       | WEIGHT |
|---|---------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | <b>Functional Coverage</b> — percentage of Must Have requirements met natively (no customization)                               | 30%    |
| 2 | <b>Scalability &amp; Performance</b> — proven ability to support 5,000+ nodes and 100,000+ interfaces                           | 15%    |
| 3 | <b>Security &amp; Compliance</b> — encryption, RBAC, audit logging, NCA ECC alignment, ISO 27001                                | 15%    |
| 4 | <b>Total Cost of Ownership (3-year)</b> — license, maintenance, support, and infrastructure costs                               | 15%    |
| 5 | <b>Vendor Stability &amp; Roadmap</b> — years in market, customer base size, active development commitment                      | 10%    |
| 6 | <b>Migration &amp; Transition Ease</b> — tooling for device discovery import, configuration migration, and parallel-run support | 8%     |
| 7 | <b>Support Quality</b> — SLA tiers available, KSA or MENA region support presence, dedicated TAM                                | 7%     |

11.1

Shortlisting Process

**Step 1 — RFI Review**  
Proposals evaluated against FR/NFR requirements matrix. Vendors failing Must Have items are disqualified.

**Step 2 — Reference Check**  
Minimum 2 enterprise references (≥1,000 nodes). GCC or healthcare sector references preferred.

**Step 3 — POC (max 3 vendors)**  
Shortlisted vendors conduct a 30-day on-site or remote POC against a defined test plan.

**Step 4 — Commercial Award**  
Final pricing submitted after POC conclusion. Contract awarded within 2 weeks of commercial close.

12

SECTION 12

Commercial & Licensing Requirements

Vendors must address all of the following in their proposal response:

| REQUIREMENT         | DETAILS                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------|
| Licensing Model     | State clearly: perpetual vs. subscription; per-node, per-interface, per-user, or unlimited              |
| 3-Year TCO          | Full 3-year cost breakdown: license fees, annual maintenance/support, implementation services, training |
| POC Pricing         | Confirm whether a fully-featured POC license can be provided at no cost during the evaluation period    |
| Scalability Pricing | Clarify pricing for starting node count (~1,000) and scaling path to 5,000+ nodes                       |
| Module Pricing      | If flow analysis, topology, or reporting are separate modules, itemize each cost separately             |
| Support Tiers       | Describe available support tiers (business hours, 24x7, dedicated TAM) and associated annual cost       |
| Payment Terms       | State preferred terms: annual, multi-year upfront; specify any multi-year discount                      |
| KSA Partner         | Confirm whether a KSA-based authorized reseller or implementation partner is available                  |

13

SECTION 13

Implementation & Support Requirements

- 13.1Implementation
- Formal implementation project plan within 10 business days of contract award
  - Parallel running period of **minimum 30 days** alongside our existing monitoring solution during cutover
  - Initial device discovery and onboarding support for the first 500 devices
  - All documentation (installation, admin, user guides) provided in English
- 13.2Training
- Administrator training: minimum **2 days** live instructor-led (in-person or virtual)
  - NOC operator training: minimum **1 day** practical session
  - Training materials (slides, video, or e-learning) for self-paced refresher use

- 13.3Ongoing Support SLA
- | PRIORITY      | INITIAL RESPONSE            |
|---------------|-----------------------------|
| P1 — Critical | ≤ 4 hours · 24x7            |
| P2 — High     | ≤ 8 hours · business hours  |
| P3 — Medium   | ≤ 24 hours · business hours |
- Dedicated support channel: ticketing portal, email, and phone
  - Security patches delivered within 30 days of release
  - Named KSA or MENA region technical account contact preferred

14

SECTION 14

Acceptance Criteria

The solution will be accepted for production when all of the following are verified through User Acceptance Testing (UAT):

| ID    | ACCEPTANCE CRITERION                                                                                       | TEST METHOD      |
|-------|------------------------------------------------------------------------------------------------------------|------------------|
| AC-01 | Discovery scan completes across a /24 subnet and returns all live devices within 5 minutes                 | Live test        |
| AC-02 | SNMP polling accurately reflects actual CPU and interface state within 2 polling cycles of a real change   | Lab test         |
| AC-03 | Alert fires and email notification is received within 60 seconds of a threshold breach                     | Live test        |
| AC-04 | Topology map correctly displays all discovered CDP/LLDP neighbors and links for a 50-device test network   | UAT review       |
| AC-05 | NetFlow data from a Cisco router correctly populates top-talker and top-application views                  | Lab test         |
| AC-06 | A Read-Only user cannot access configuration or alert management screens                                   | Role test        |
| AC-07 | A scheduled PDF report is generated and delivered to the configured email recipient                        | End-to-end test  |
| AC-08 | Platform sustains polling of 10,000 interfaces for 24 hours without memory leak or performance degradation | Load test        |
| AC-09 | HA failover completes within 60 seconds with no data loss on the secondary node                            | Failover test    |
| AC-10 | RBAC user scoped to a device group cannot view devices outside that group                                  | Role test        |
| AC-11 | Alert suppression correctly silences child-device alerts when the upstream parent is confirmed down        | Integration test |
| AC-12 | Audit log captures login, configuration change, and alert acknowledgement with user and timestamp          | Audit test       |

15

SECTION 15

Risks & Constraints

| RISK                                                    | LIKELIHOOD | IMPACT    | MITIGATION                                                                           |
|---------------------------------------------------------|------------|-----------|--------------------------------------------------------------------------------------|
| Monitoring gap during platform cutover                  | Medium     | High      | Mandatory 30-day parallel-run; vendor to confirm migration tooling                   |
| MIB compatibility issues with niche medical devices     | Medium     | Medium    | Vendor must provide MIB browser, OID test tool, and escalation path for custom MIBs  |
| Flow data volume exceeding platform processing capacity | Low        | High      | Vendor to provide capacity sizing guidance; confirm flow sampling support            |
| Database storage growth exceeding projections           | Medium     | High      | Vendor must provide configurable retention policies and auto-archive capability      |
| ServiceNow integration requiring custom development     | Low        | Medium    | Confirm native or documented REST API integration before contract award              |
| Vendor financial instability or product discontinuation | Low        | Very High | Require 3-year roadmap commitment; evaluate market standing and active customer base |

16

SECTION 16

Vendor Response Instructions

16.1

Required Sections in Vendor Proposal

| #  | SECTION                         | CONTENT EXPECTED                                                                                                                                                                                    |
|----|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Company Overview                | Years in operation, GCC customer base, relevant certifications (ISO 27001, etc.)                                                                                                                    |
| 2  | Product Overview                | Current product version, release cadence, 3-year roadmap summary                                                                                                                                    |
| 3  | Requirements Compliance Matrix  | Complete table mapping every FR and NFR in this document to: <span>✔</span> Natively supported / <span>⚠</span> Via configuration / <span>🔧</span> Via customization / <span>✖</span> Not supported |
| 4  | Architecture Overview           | Deployment diagram, HA design, supported OS, virtualization platforms, storage sizing                                                                                                               |
| 5  | Security & Compliance Statement | Explicitly address NCA ECC 2018, encryption at rest/in-transit, audit logging, data residency                                                                                                       |
| 6  | Migration Approach              | Proposed transition plan with zero monitoring gap; parallel-run strategy and cutover approach                                                                                                       |
| 7  | POC Plan                        | Proposed 30-day POC scope, success criteria, and resource requirements                                                                                                                              |
| 8  | Customer References             | Minimum 2 enterprise references; GCC or healthcare sector preferred                                                                                                                                 |
| 9  | Commercial Proposal             | Itemized 3-year TCO, licensing model, support tiers, and payment options                                                                                                                            |
| 10 | Support Model                   | SLA commitments, escalation process, KSA/MENA region support availability                                                                                                                           |

16.2

Submission Details

|                                                                                                           |                                                                                            |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Submission Format</b><br>PDF or Word document submitted via email                                      | <b>Point of Contact</b><br>Procurement Office<br>shabeeb.k@code-ox.com<br>+966 535 716 437 |
| <b>Questions Deadline</b><br>Clarifications accepted up to 5 business days before the submission deadline | <b>Submission Deadline</b><br>To be confirmed upon BRD distribution to shortlisted vendors |

A

APPENDIX A

Priority Key

| PRIORITY    | DEFINITION                                                                                 | VENDOR IMPLICATION                                                       |
|-------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Must Have   | Critical requirement — the solution must support this natively or with minor configuration | Failure to meet a Must Have item <b>disqualifies</b> the vendor proposal |
| Should Have | Important but not disqualifying — absence will be scored against the vendor                | Describe compensating capability or planned roadmap delivery date        |
| Could Have  | Nice to have — low weight in scoring; not expected in baseline proposal                    | Mention if natively available; no penalty if absent                      |

B

APPENDIX B

Glossary

| TERM    | DEFINITION                                                                         |
|---------|------------------------------------------------------------------------------------|
| SNMP    | Simple Network Management Protocol — polls and manages network devices             |
| NetFlow | Cisco protocol for exporting IP traffic metadata from routers and switches         |
| sFlow   | Vendor-neutral flow sampling protocol supported by HPE/Aruba and others            |
| IPFIX   | IETF-standardized flow export protocol (successor to NetFlow v9)                   |
| CDP     | Cisco Discovery Protocol — Layer 2 protocol for discovering adjacent Cisco devices |
| LLDP    | Link Layer Discovery Protocol — vendor-neutral equivalent of CDP                   |
| MIB     | Management Information Base — SNMP OID database for a given device type            |
| MTTR    | Mean Time to Repair — average time to restore a failed service or device           |
| MTBF    | Mean Time Between Failures — average time between successive failures              |

| TERM    | DEFINITION                                                                       |
|---------|----------------------------------------------------------------------------------|
| RBAC    | Role-Based Access Control — permission model tying user roles to resource access |
| NCA ECC | Saudi Arabia National Cybersecurity Authority Essential Controls (2018)          |
| NOC     | Network Operations Center — team responsible for real-time network monitoring    |
| HA      | High Availability — system design eliminating single points of failure           |
| TCO     | Total Cost of Ownership — all costs over a defined ownership period              |
| POC     | Proof of Concept — time-limited evaluation of a vendor solution                  |
| ITSM    | IT Service Management platform (e.g., ServiceNow)                                |
| TLS     | Transport Layer Security — protocol for encrypted data transmission              |
| SAML    | Security Assertion Markup Language — standard for SSO federation                 |